SIMATS SCHOOL OF ENGINEERING

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| Course Code: DSA03 | Course Name: Natural Language Processing |
|---|---|
| Branch: CSE | Year : 2026 |

**Name: R.Rohith Manikanta**
**Reg no: 192425326**

**Assignment 3**

**1. Draw tree structures for the following ATIS sentences:** (10M) (CO3) (BLT4)

**a. Does American airlines have a flight between five a.m. and six a.m.**
**b. I would like to fly on American airlines.**
**c. Please repeat that.**
**d. Does American 487 have a first class section?**
**e. I need to fly between Philadelphia and Atlanta.**
**f. What is the fare from Atlanta to Denver?**
**g. Is there an American airlines flight from Philadelphia to Dallas?**

**2. Consider the below grammar to construct the Early parser for the input string "Christ Opened the eyes".** (10M) (CO3) (BLT4)

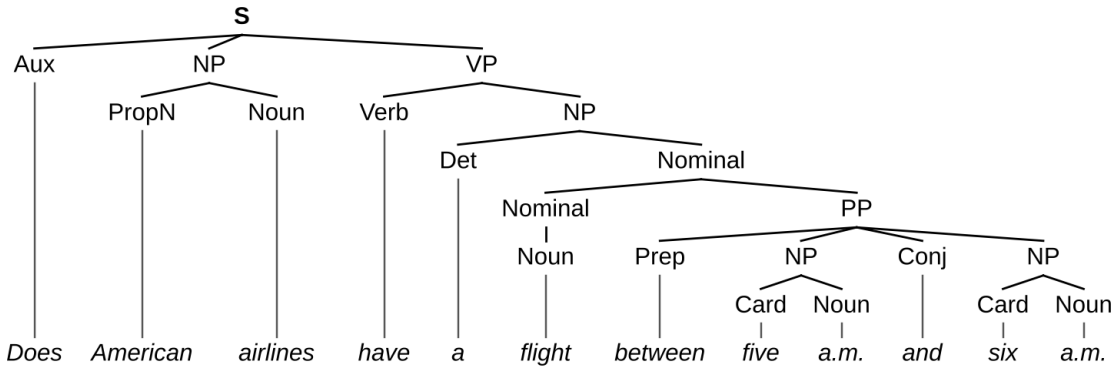
**S -> NP VP VP -> V NP PP V -> opened**
**NP -> N PP -> Prep NP V -> ate**
**NP -> Det NP N -> Christ Det -> the**
**NP -> NP NP N -> blind Det -> a**
**VP -> V N -> eyes Prep -> of**

**VP -> V NP N -> bread Prep -> in**

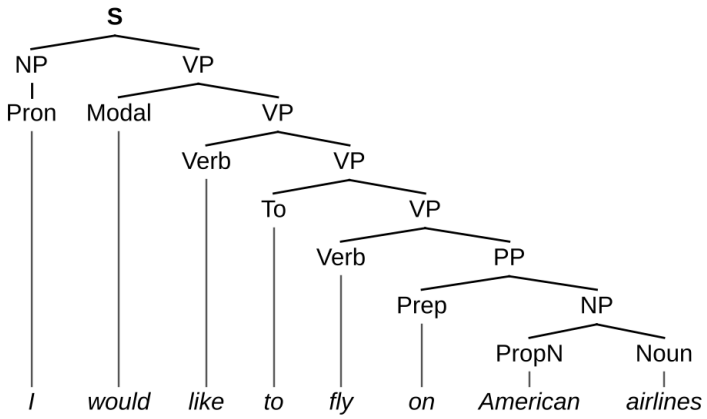
## Answer 1. Tree structures for the ATIS sentences

The trees use a small ATIS-style grammar: S → NP VP | Aux NP VP | VP | NP VP | Aux NP NP; NP → Pron | PropN Noun | PropN Card | Det Nominal | NP-conjoined PP; Nominal → Noun | Adj Nominal | Nominal Noun | Nominal PP; VP → Verb NP | Modal VP | To VP | Verb VP | Verb PP | Adv VP; PP → Prep NP (Conj NP). Labels: Aux = auxiliary, PropN = proper noun, Pron = pronoun, Det = determiner, Card = cardinal number, Conj = conjunction, Adv = adverb, WhPron = wh-pronoun, To = infinitive "to". In each tree the sentence words are shown in italics along the bottom.

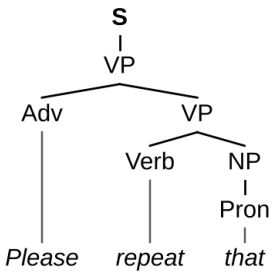
**a. Does American airlines have a flight between five a.m. and six a.m.**



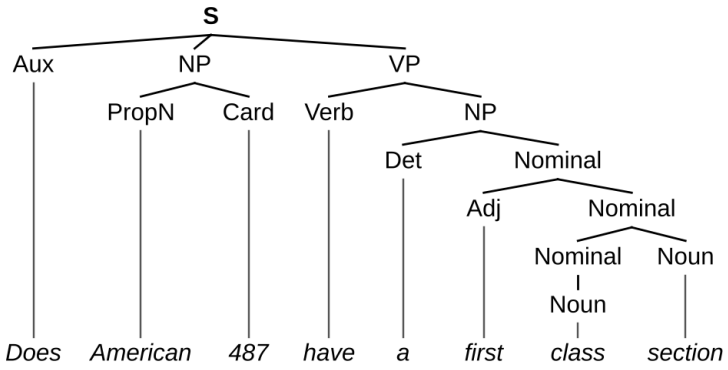

**b. I would like to fly on American airlines.**



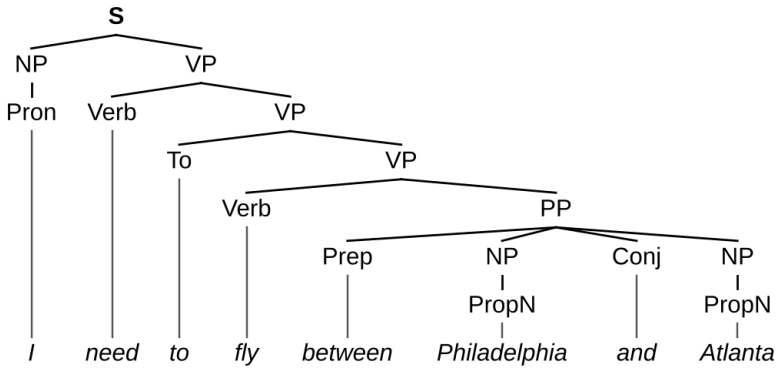

**c. Please repeat that.**



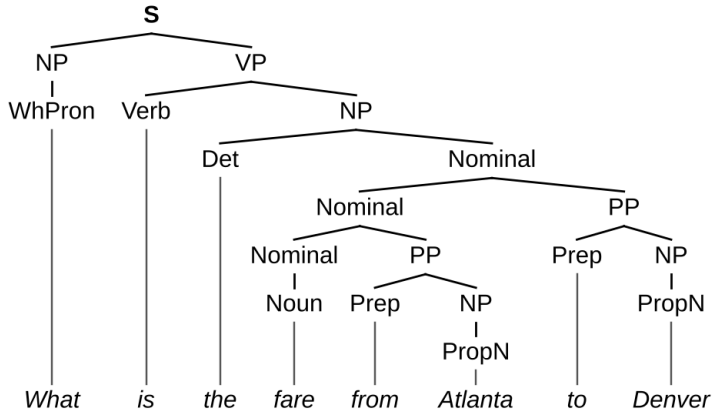
**d. Does American 487 have a first class section?**



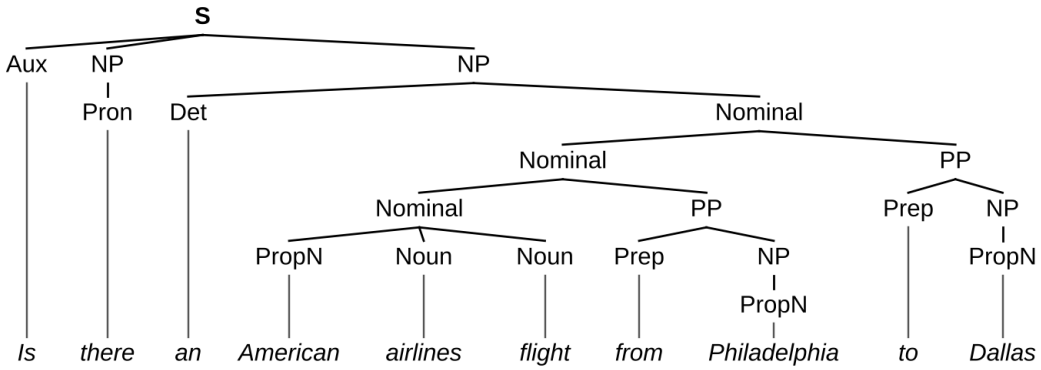

**e. I need to fly between Philadelphia and Atlanta.**




**f. What is the fare from Atlanta to Denver?**



**g. Is there an American airlines flight from Philadelphia to Dallas?**




## Answer 2. Earley parse of “Christ opened the eyes”

**Grammar read from the question:** S → NP VP; NP → N | Det NP | NP NP; VP → V | V NP | V NP PP; PP → Prep NP; V → opened | ate; N → Christ | blind | eyes | bread; Det → the | a; Prep → of | in. The input is normalised to lower case (“Christ opened the eyes”), so positions are: 0 Christ 1 opened 2 the 3 eyes 4. A state is written (rule with • marking progress, [i, j]) where i is where the constituent starts and j is the position of the dot. The start state γ → • S is added to chart S0, and the parse succeeds if γ → S • [0,4] appears in the last chart.

**Chart S0**

| State | Rule | [i, j] | Operation |
|---|---|---|---|
| S0.0 | γ → • S | [0, 0] | Start |
| S0.1 | S → • NP VP | [0, 0] | Predictor |
| S0.2 | NP → • N | [0, 0] | Predictor |
| S0.3 | NP → • Det NP | [0, 0] | Predictor |
| S0.4 | NP → • NP NP | [0, 0] | Predictor |

**Chart S1 (after reading “Christ”)**

| State | Rule | [i, j] | Operation |
|---|---|---|---|
| S1.0 | N → christ • | [0, 1] | Scanner (word “christ”) |
| S1.1 | NP → N • | [0, 1] | Completer |
| S1.2 | S → NP • VP | [0, 1] | Completer |
| S1.3 | NP → NP • NP | [0, 1] | Completer |
| S1.4 | VP → • V | [1, 1] | Predictor |
| S1.5 | VP → • V NP | [1, 1] | Predictor |
| S1.6 | VP → • V NP PP | [1, 1] | Predictor |
| S1.7 | NP → • N | [1, 1] | Predictor |
| S1.8 | NP → • Det NP | [1, 1] | Predictor |
| S1.9 | NP → • NP NP | [1, 1] | Predictor |

**Chart S2 (after reading “opened”)**

| State | Rule | [i, j] | Operation |
|---|---|---|---|
| S2.0 | V → opened • | [1, 2] | Scanner (word “opened”) |
| S2.1 | VP → V • | [1, 2] | Completer |
| S2.2 | VP → V • NP | [1, 2] | Completer |
| S2.3 | VP → V • NP PP | [1, 2] | Completer |
| S2.4 | S → NP VP • | [0, 2] | Completer |
| S2.5 | NP → • N | [2, 2] | Predictor |
| S2.6 | NP → • Det NP | [2, 2] | Predictor |
| S2.7 | NP → • NP NP | [2, 2] | Predictor |
| S2.8 | γ → S • | [0, 2] | Completer |

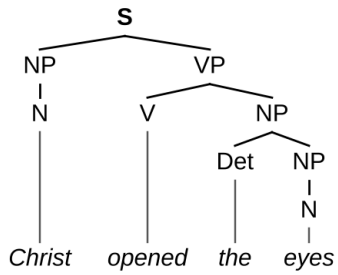
**Chart S3 (after reading “the”)**

| State | Rule | [i, j] | Operation |
|---|---|---|---|
| S3.0 | Det → the • | [2, 3] | Scanner (word “the”) |
| S3.1 | NP → Det • NP | [2, 3] | Completer |
| S3.2 | NP → • N | [3, 3] | Predictor |
| S3.3 | NP → • Det NP | [3, 3] | Predictor |
| S3.4 | NP → • NP NP | [3, 3] | Predictor |

**Chart S4 (after reading “eyes”)**

| State | Rule | [i, j] | Operation |
|---|---|---|---|
| S4.0 | N → eyes • | [3, 4] | Scanner (word “eyes”) |
| S4.1 | NP → N • | [3, 4] | Completer |
| S4.2 | NP → Det NP • | [2, 4] | Completer |
| S4.3 | NP → NP • NP | [3, 4] | Completer |
| S4.4 | VP → V NP • | [1, 4] | Completer |
| S4.5 | VP → V NP • PP | [1, 4] | Completer |
| S4.6 | NP → NP • NP | [2, 4] | Completer |
| S4.7 | NP → • N | [4, 4] | Predictor |
| S4.8 | NP → • Det NP | [4, 4] | Predictor |
| S4.9 | NP → • NP NP | [4, 4] | Predictor |
| S4.10 | S → NP VP • | [0, 4] | Completer |
| S4.11 | PP → • Prep NP | [4, 4] | Predictor |
| S4.12 | γ → S • | [0, 4] | Completer |

**Result.** State S4.12, γ → S • [0,4], is in the final chart, so the string is accepted. Following the completer links back gives the parse S[0,4] → NP[0,1] VP[1,4], with NP → N (Christ) and VP → V NP, where V = opened and NP[2,4] → Det NP (the + NP → N eyes). The other states in the charts (for example NP → NP • NP) are partial hypotheses created by the left-recursive rule NP → NP NP that never complete, so the sentence has exactly one parse.



# Supplementary Explanation - Answer 1

## Reading the ATIS constituency trees

A constituency tree shows how a sentence is grouped into phrases. The top node, usually S, represents the complete sentence. Branches below it represent larger constituents such as noun phrases (NP), verb phrases (VP), and prepositional phrases (PP). The words at the bottom are the terminal items of the tree. Reading upward from a word shows the phrase to which it belongs; reading downward from S shows how the whole sentence is assembled.

*The trees on pages 2-4 are retained as supplied. The notes below explain their structures and the function of the main labels without changing the diagrams.*

## Common labels used in the trees

| Label | Meaning | Role in the ATIS examples |
|---|---|---|
| S | Sentence | Top-level constituent covering the complete utterance. |
| NP | Noun phrase | Names an entity, person, airline, route point, object, or pronoun. |
| VP | Verb phrase | Contains the main verb and the complements or modifiers associated with it. |
| PP | Prepositional phrase | Begins with a preposition such as on, between, from, or to. |
| Aux / Modal | Auxiliary / modal verb | Marks a question or modality, as in Does, Is, and would. |
| Det / Adj | Determiner / adjective | Introduces or describes a noun, as in a, the, first, and class. |
| PropN / Card | Proper noun / cardinal | Identifies a named airline or city, or a flight number such as 487. |
| Conj / Adv | Conjunction / adverb | Joins coordinated phrases or modifies an utterance, as in and or Please. |

## Sentence interpretations: (a)-(c)

| Sentence | Main grouping in the supplied tree | What the structure tells us |
|---|---|---|
| (a) Does American airlines have a flight between five a.m. and six a.m.? | Aux + NP + VP | Does is placed before the subject NP. The VP contains have and the object NP a flight. The time expression is represented as a PP with two coordinated time expressions joined by and. |
| (b) I would like to fly on American airlines. | NP + VP | I is the subject NP. The VP contains the modal would, the verb like, the infinitival marker to, and the verb fly. The PP on American airlines supplies the airline information associated with flying. |
| (c) Please repeat that. | VP (imperative) | The sentence is an instruction. Please is shown as an adverbial element, repeat is the main verb, and that is the NP complement of repeat. An overt subject is not shown in the supplied tree. |

The main benefit of the tree representation is that it distinguishes a word sequence from its grammatical grouping. For example, between five a.m. and six a.m. is interpreted as one coordinated time expression within the larger object phrase, rather than as unrelated words.

# Supplementary Explanation - Answer 1 (continued)

## Sentence interpretations: (d)-(g)

| Sentence | Main grouping in the supplied tree | What the structure tells us |
|---|---|---|
| (d) Does American 487 have a first class section? | Aux + NP + VP | Does marks the yes/no question. American 487 forms the subject NP, with American as a proper name and 487 as a flight-number/cardinal element. The VP contains have and the object NP a first class section; first and class modify the head noun section. |
| (e) I need to fly between Philadelphia and Atlanta. | NP + VP | I is the subject. The VP is built around need, followed by the infinitival sequence to fly. The PP between Philadelphia and Atlanta expresses a route relation and contains two coordinated city NPs. |
| (f) What is the fare from Atlanta to Denver? | Wh element + VP + NP | What introduces the information request. The NP the fare is expanded by location PPs. From Atlanta and to Denver identify the route endpoints associated with the fare. The supplied tree uses nested nominal and PP groupings to attach these phrases. |
| (g) Is there an American airlines flight from Philadelphia to Dallas? | Aux + NP + NP | Is is the auxiliary at the beginning of the yes/no question. The supplied tree places there in an NP position and groups an American airlines flight as the following NP. The PP from Philadelphia to Dallas specifies the route of that flight. |

## Patterns shared by the seven sentences

Question formation. Sentences (a), (d), and (g) are yes/no questions. Their trees place an auxiliary at the beginning, before the subject-related material. Sentence (f) is a wh-question: the wh-word What occurs at the front because the speaker is asking for a particular item of information. Sentence (c) is an imperative, so the supplied tree begins directly with a VP.

Prepositional phrases. The examples contain several PPs: on American airlines, between Philadelphia and Atlanta, from Atlanta, to Denver, and from Philadelphia to Dallas. A PP normally consists of a preposition followed by an NP. Its interpretation depends on the phrase it attaches to: it may identify an airline, a route, a time interval, or other information.

Coordination. The conjunction and joins items that have a parallel role. In (a), it joins the two time expressions. In (e), it joins the two city names. This grouping is useful because the tree represents the coordinated sequence as a unit within the larger phrase.

Proper names and numbers. Airline names and city names are treated as proper-name material in the supplied trees. The number 487 is a flight identifier. Keeping these items inside their appropriate NPs helps distinguish names and identifiers from ordinary nouns.

> *These are task-oriented ATIS trees, not a complete grammar of English. They provide a compact representation of the structures needed for the seven travel-query examples. The original diagrams remain the authoritative visual representation for each sentence.*

# Supplementary Explanation - Answer 2

## The Earley parser: purpose and state notation

The second question asks for an Earley parse of the input “Christ opened the eyes” using the grammar printed in the assignment. The Earley algorithm is a chart-parsing method: it stores partial grammatical hypotheses at each input position, so it does not need to restart the analysis for every possible phrase grouping.

The input is treated as four tokens. The chart position is a boundary between tokens, so position 0 is before the first word and position 4 is after the last word.

| Boundary position | Token between this boundary and the next |
|---|---|
| 0 to 1 | Christ |
| 1 to 2 | opened |
| 2 to 3 | the |
| 3 to 4 | eyes |

A chart state is written as a grammar rule with a dot and a span [i, j]. The dot marks how much of the rule has been recognized. The left number i is the starting input position of the constituent; j is the current position reached by the parser. For example, NP → Det • NP [2, 3] means that an NP began at position 2, the determiner has been recognized through position 3, and another NP is expected next.

## Three operations build the chart

| Operation | When it is used | Effect |
|---|---|---|
| Predictor | The symbol immediately after the dot is a nonterminal. | Adds the grammar rules that could expand that nonterminal at the current position. |
| Scanner | The symbol immediately after the dot is a terminal word that matches the next input token. | Consumes that word and advances the dot and the right boundary by one position. |
| Completer | A state has finished recognizing a constituent. | Finds earlier states that were waiting for this constituent and advances their dots. |

## Initial chart: S0

The augmented start rule γ → • S is inserted first. Since the dot is before S, the Predictor adds the production S → NP VP. It then predicts the possible NP expansions at position 0: NP → N, NP → Det NP, and NP → NP NP. These states are hypotheses; none of them has consumed a word yet.

## Expanded reading of the initial chart

| State | Rule | [i, j] | Why it appears |
|---|---|---|---|
| S0.0 | γ → • S | [0, 0] | Introduces the augmented start symbol and requests a complete S. |
| S0.1 | S → • NP VP | [0, 0] | Predicts the assignment grammar rule for a sentence. |
| S0.2 | NP → • N | [0, 0] | An NP may be formed directly from a noun. |
| S0.3 | NP → • Det NP | [0, 0] | An NP may begin with a determiner followed by another NP. |
| S0.4 | NP → • NP NP | [0, 0] | The left-recursive NP-combination rule is also considered. |

## After scanning “Christ” (chart S1)

The first token matches N → Christ. The Scanner advances that lexical rule to a completed state over [0, 1]. The Completer can then finish NP → N and advance S → NP • VP, because the subject NP has been recognized. The parser predicts VP expansions at position 1, while the left-recursive NP rule also creates additional hypotheses.

# Supplementary Explanation - Answer 2 (continued)

## Expanded chart reading: S1

| State | Rule | [i, j] | Interpretation |
|---|---|---|---|
| S1.0 | N → christ • | [0, 1] | The word Christ has matched a noun. |
| S1.1 | NP → N • | [0, 1] | The noun completes a noun phrase. |
| S1.2 | S → NP • VP | [0, 1] | The subject NP is complete; a VP is still required. |
| S1.3 | NP → NP • NP | [0, 1] | A second NP could extend the NP under the recursive rule. |
| S1.4 | VP → • V | [1, 1] | Predicts a VP consisting of a verb. |
| S1.5 | VP → • V NP | [1, 1] | Predicts a verb followed by an NP. |
| S1.6 | VP → • V NP PP | [1, 1] | Predicts a verb, an NP, and a PP. |
| S1.7 | NP → • N | [1, 1] | Predicts noun material if an NP is needed at position 1. |
| S1.8 | NP → • Det NP | [1, 1] | Predicts a determiner-led NP at position 1. |
| S1.9 | NP → • NP NP | [1, 1] | Retains the recursive NP possibility at position 1. |

## After scanning “opened” (chart S2)

At position 1, the input word opened matches the lexical rule V → opened. That completed V can satisfy the beginning of several VP rules. In the supplied chart, VP → V • is complete, while VP → V • NP and VP → V • NP PP remain active because they expect more material. The latter two possibilities are important for recognizing the object NP “the eyes”.

## Expanded reading of chart S2

| State | Rule | [i, j] | Interpretation |
|---|---|---|---|
| S2.0 | V → opened • | [1, 2] | The second token has been recognized as a verb. |
| S2.1 | VP → V • | [1, 2] | A verb-only VP is complete under the supplied rule set. |
| S2.2 | VP → V • NP | [1, 2] | A VP can continue with an object NP. |
| S2.3 | VP → V • NP PP | [1, 2] | A VP can continue with an NP and then a PP. |
| S2.4 | S → NP VP • | [0, 2] | The chart records a sentence-sized constituent over the prefix Christ opened. |
| S2.5 | NP → • N | [2, 2] | Predicts a noun-led object NP at the current position. |
| S2.6 | NP → • Det NP | [2, 2] | Predicts a determiner-led object NP; this matches the next token the. |
| S2.7 | NP → • NP NP | [2, 2] | Keeps the recursive NP alternative available. |
| S2.8 | γ → S • | [0, 2] | The augmented rule records completion for the prefix span [0, 2]. |

*The S2 chart contains a complete analysis for the prefix “Christ opened” because the supplied grammar includes a verb-only VP option. The full assignment input continues with “the eyes”; therefore, the requested full-span derivation is checked at the final boundary 4, as shown in chart S4 and the result on the original page 5.*

## After scanning “the” (chart S3)

The token the matches Det → the. This completes the determiner portion of NP → Det NP over [2, 3]. The parser then predicts possible NP continuations beginning at position 3. The next word, eyes, can match the noun rule and complete the expected NP.

# Supplementary Explanation - Answer 2 (continued)

## Final chart: after scanning “eyes” (S4)

The final token eyes matches N → eyes over [3, 4]. This completes NP → N for the final noun phrase. That NP completes the determiner phrase NP → Det NP over [2, 4]. The resulting object NP then completes VP → V NP over [1, 4]. Finally, the VP combines with the subject NP over [0, 1] to complete S → NP VP over the full input span [0, 4].

| State | Rule | [i, j] | Interpretation |
|---|---|---|---|
| S4.0 | N → eyes • | [3, 4] | The final word is recognized as a noun. |
| S4.1 | NP → N • | [3, 4] | The noun eyes completes an NP. |
| S4.2 | NP → Det NP • | [2, 4] | The sequence the eyes completes the determiner-led NP. |
| S4.3 | NP → NP • NP | [3, 4] | A recursive NP continuation is recorded. |
| S4.4 | VP → V NP • | [1, 4] | Opened plus the object NP completes a VP. |
| S4.5 | VP → V NP • PP | [1, 4] | A VP with an optional following PP remains possible under the rule. |
| S4.6 | NP → NP • NP | [2, 4] | The completed NP may be used by the recursive NP rule. |
| S4.7 | NP → • N | [4, 4] | Predicts a noun-led NP at the end boundary. |
| S4.8 | NP → • Det NP | [4, 4] | Predicts a determiner-led NP at the end boundary. |
| S4.9 | NP → • NP NP | [4, 4] | Retains the recursive NP option at the end boundary. |
| S4.10 | S → NP VP • | [0, 4] | The subject and VP form a sentence spanning the entire input. |
| S4.11 | PP → • Prep NP | [4, 4] | Predicts a PP possibility at the end boundary. |
| S4.12 | γ → S • | [0, 4] | The augmented start rule is complete over all four tokens. |

## Completed derivation for the requested sentence

Step 1 - Subject NP: Christ is recognized as N, and N expands to NP using NP → N. This gives NP[0,1].

Step 2 - Main verb: opened is recognized by V → opened. The verb is the first part of the VP.

Step 3 - Object NP: the is recognized as Det and eyes as N. The noun forms NP → N, and the determiner combines with it using NP → Det NP. This gives NP[2,4].

Step 4 - Verb phrase: the verb and its object combine through VP → V NP, giving VP[1,4].

Step 5 - Sentence: the subject NP and VP combine through S → NP VP, giving S[0,4]. The augmented state γ → S • [0,4] confirms that the complete four-token input has been recognized.

## Final bracketed structure

*S [0,4] → NP [0,1] VP [1,4]*
*NP [0,1] → N [0,1] → Christ*
*VP [1,4] → V [1,2] NP [2,4]*
*V [1,2] → opened*
*NP [2,4] → Det [2,3] NP [3,4] → the eyes*

## Conclusion

The Earley chart shows how a parser can combine lexical matches, partial phrases, and completed constituents while moving from left to right through a sentence. Predictor creates possible continuations, Scanner matches the next input word, and Completer joins newly completed constituents to earlier states. For “Christ opened the eyes”, the final state γ → S • [0,4] shows that the whole input is accepted. The final parse tree is retained on the original page 5.

*Together, the original tree diagrams and chart tables, plus these added explanations, form a ten-page assignment. The original question sheet and all original diagrams and chart tables have been retained.*